

RISKA DERLIANA PURBA

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EDUCATION

Politeknik Negeri Batam - Batam, Indonesia

Bachelor of Applied Mechatronics Engineer | 3.91 GPA

09/2020 to 04/2024

JOB EXPERIENCE

Metronarc Technology Pte. Ltd. - Batam, Indonesia

Product Development Engineer

04/2024 to present

- Designed and developed mechanical enclosures using SolidWorks, ensuring compatibility with diverse client machinery.
- Created detailed assembly drawings for seamless device integration, adhering to technical standards.
- Analyzed client needs and improved designs to address varying machine specifications.
- Designed and optimized PCB modules, integrating hardware and software for peak performance.
- Led prototype testing and troubleshooting during installation to identify and resolve technical issues efficiently.
- Validated product performance through iterative testing and user-driven design enhancements.
- Collaborated with teams to communicate product features, boosting market competitiveness.

Sumitomo Wiring System Batam Indonesia - Batam, Indonesia

Assembly Improvement Analyst Intern

07/2023 to 03/2024

- Improved production line efficiency by 5% through process optimization and a 3% daily reduction in downtime.
- Conducted root cause analysis to resolve production defects and equipment downtime, ensuring issue prevention.
- Implemented process enhancements in manpower allocation, machine performance, and workflow, boosting productivity.
- Monitored efficiency metrics to drive continuous improvements and operational excellence.

AWARDS

National Competition E-TIME, 7th Semester, 2023

Competed in the Programmable Logic Controller (PLC) category using the TM221 Schneider PLC model at Jakarta State Polytechnic.

35th Pekan Ilmiah Mahasiswa Nasional (PIMNAS), 5th Semester, 2022

Represented in the Karsa Cipta (PKM-KC) category at Universitas Muhammadiyah Malang with an IoT-based innovative project.

Winner of Project Based Learning (PBL) EXPO, 5th Semester, 2022

Achieved First Place in the Internet of Things category at the PBL EXPO organized by Politeknik Negeri Batam.

Funding Recipient Program Kreatifitas Mahasiswa (PKM)-KC, 5th Semester, 2022

Secured funding in the Karsa Cipta category for the project "Rawai Pintar: A Smart Longline System to Detect Fish Bites and Location with Time and Fuel Efficiency, Integrated with Smartphones".

Funding Recipient Program Kreatifitas Mahasiswa (PKM), 5th Semester, 2022

Secured funding in the Karya Inovatif category for the project "IoT-Based Baby Weighing Scale for Early Detection of Stunting Using Z-Score Methodology".

Outstanding Student of Batam Polytechnic, 5th Semester, 2022

Recognized as an Outstanding Student for advancing in the PKM National Program and PIMNAS under the innovative work category.

PROJECT EXPERIENCE

SPARK “System Monitoring Machine Welding”

Role: LoRa System Integration, Mechanical Design, and UI Development

10/2022 to 06/2023

- Implemented LoRa communication for stable, long-range data transmission.
- Designed mechanical structure using SolidWorks for portability and easy installation.
- Developed an intuitive web interface, improving user engagement and navigation.

IoT-Based Baby Weighing Scale

Role: Sensor Calibration, Data Transmission, and Electrical Wiring

10/2021 to 08/2022

- Calibrated load cell, ultrasonic, and rotary sensors for precise measurements.
- Integrated ESP32 microcontroller with Firebase for real-time data transmission and monitoring.
- Designed and executed stable electrical wiring for field durability.

Water pH Detection Tool for Fish Ponds

Role: Sensor Calibration and Electrical Wiring

01/2021 to 08/2021

- Calibrated pH sensor for reliable water quality monitoring.
- Designed electrical wiring and component integration for consistent performance.

CERTIFICATIONS

Programmable Logic Controller – Badan Nasional Sertifikasi Profesi (BNSP)

11/2023 to 11/2026

Credential ID 26510311400000112023

Competence SMEs and Entrepreneurs – Badan Nasional Sertifikasi Profesi (BNSP)

12/2022 to 12/2025

Credential ID 47000122000219982022

Learn Schneider Plc Programming with TM221 – Udemy

06/2023

Credential ID UC-f6260e24-76bc-4707-8c26-890ea4bf043e

SKILLS

Mechanical Design: Proficient in designing and developing mechanical enclosures and components using SolidWorks.

PCB Design: Skilled in creating and optimizing PCB layouts using EAGLE.

PLC Programming: Experienced with Schneider EcoStruxure Machine Expert and CX-One programming.

Electrical Components and Wiring: Knowledgeable in electrical components, symbol reading, and wiring system implementation.

Troubleshooting and Problem-Solving: Adept at diagnosing and resolving mechanical, electrical, and software issues

Technical Drawing: Competent in designing schematic diagrams and electrical drawings for system implementation.

Analytical Skills: Strong at analyzing client needs and optimizing designs for better performance.

ORGANIZATION EXPERIENCE

Himpunan Mahasiswa Elektro (HME)

03/2021 to 01/2022

Member of Departement Skill and Quality

- Committee for the "Pengaderan Elektro" 2021
- Committee for the Robotic Contest 2021